The Economics of Sharding

Which markets make a shard pay — and in what order

When a F1R3FLY shard pays for itself, every target sector becomes viable. The strategy is not *whether* to serve them, but the sequence that hardens the platform while it grows.

0.2–4%
shard load to break even, in every revenue-bearing sector

$0.00014
network cost per data payload at dense packing

1 : 20
one ad impression covers ∼20 data payloads

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Companion to the working notes *Rent and Shard Splitting* and *Sector Economics and Go-to-Market Viability*.

Executive summary

F1R3FLY's per-shard cost is small and its dense-packing dispatch (built on the 2025 elastic-hashing result) is cheap, so a single ten-node shard — about $4,160/month — breaks even at a *tiny fraction* of its capacity in every sector that bears a transaction fee. Across AI, fintech, biotech, medical, and music, that fraction runs from one-fifth of one percent to four percent. **Capacity is never the constraint on viability; demand is.**

Three findings shape the go-to-market plan:

- **Risk pays.** Revenue density and data risk rise together, and both run opposite to volume. The riskiest data (patient records) is the *easiest* to make profitable per transaction; the cheapest, highest-volume data (consumer media) is the hardest.

- **Participatory advertising unlocks free consumer Web 2.0 — and inverts it.** Users pay their network cost in tokenised ad impressions; one impression covers roughly twenty data payloads. Past break-even they keep the surplus as portable tokens. The user stops being the product and becomes a paid participant.
- **Sequence beats selection.** The right order is AI first (anchor and growth engine), then fintech (margin), then medical (annuity), with music as continuous fill and Web 2.0 as the eventual mass-market substrate — preserving a safe-to-critical risk progression throughout.

The viability condition

A shard is viable when monthly revenue covers its cost. Break-even traffic is $T^\star = C_{\text{eff}}/r_{\text{tx}}$ — cost over fee-per-transaction. Because a shard carries on the order of $30{,}000{,}000$ transactions/month before it splits, the meaningful figure is the *break-even load*: the slice of one shard a sector must fill to pay for it.

Sector	Revenue density	Risk	Growth ceiling	Break-even	Strategic role
AI	$0.01 / tx	low–mod	none (machine vol.)	$\sim 2\%$	**anchor / growth**
Fintech	$\sim$$0.25 / tx	high	semi-bounded	$\sim 0.2\%$	margin engine
Medical	high / patient	highest	hard cohort cap	$\sim 4\%$	cash-cow annuity
Music	$0.0003 / tx	low	roster-dependent	$\sim$full	cold-tail fill
Web 2.0	ad-funded	low–mod	astronomical	~ 3 ads/day	mass-market moat

AI is the only sector with no growth ceiling: machine-generated volume is unbounded and sticky, so it both stress-tests the dispatch hot path and scales without limit. **Fintech** is the densest — it pays for a shard at one-fifth of one percent load. **Medical** is a capped annuity: serving 80% of the VA's ~ 9.1M enrolled veterans yields $\sim \$87$M/year of revenue against $\sim \$3.8$M of infrastructure, but it saturates and then plateaus, and its risk profile demands the most assurance — so it deploys last. **Music** is high-margin but roster-fragile, ideal as continuous fill rather than as an anchor.

The participatory advertising inversion

Free consumer products have no per-user fee, so they appear to break the model — until advertising is made participatory. End users pay in tokenised impressions per data payload until their usage is covered, then either switch to impression-free payloads or accumulate advertising tokens spendable in other markets.

> A data payload costs **$0.00014**. A display impression at a $\$\sim 3$ CPM is worth **$0.003**.
>
> **One ad impression covers the network cost of ~ 20 data payloads.**

Every major property is funded if a user sees just two or three ads a day — far below real exposure — so the typical user crosses into *net-earning* portable tokens. Because impressions are themselves on-chain, fee-bearing transactions, the ad layer is self-funding volume rather than overhead, and the network still takes a thin margin at vast scale (a $0.00002 take per payload across Gmail-scale volume is $\sim \$72$M/month).

This is the inversion. In incumbent Web 2.0 the user is the product: attention is extracted and the surplus is captured by the platform. Here the user covers their own cost with attention and **keeps the surplus**. Tokens earned on ad-tolerant, high-value surfaces (e.g. maps) pay for impression-free use of ad-averse ones (e.g. documents) — a cross-subsidy carried in the *user's* wallet, not wielded as the platform's lever. It is "the community is the moat" written as an income statement, and the structural answer to enshittification.

We enter not at full scale but through a **privacy wedge**: the $\sim 1\%$ of users who actively want to own their data and earn from their attention. That slice is trivially funded and self-selects exactly the early adopters the pitch is built for.

Recommended sequence

1. **AI (ASI Alliance) — anchor.** Real volume hardens the dispatch/COMM hot path; unbounded, self-funding growth.
2. **Fintech (Boring Financial) — margin.** Highest revenue density; builds high-assurance discipline on recoverable failures.
3. **Medical (VA) — annuity.** Highest per-unit margin, deployed last so the assurance and funded-floor guarantees are battle-tested first.
4. **Music — continuous fill.** Cold-tail packing that monetises spare capacity across the network.
5. **Web 2.0 — mass-market substrate.** Entered through the privacy wedge once the participatory ad rails are proven.

The original safe-to-critical risk progression is preserved; the economic ordering simply layers on top of it, putting the unbounded, self-funding sector first and the highest-assurance sector last.

All figures are engineering estimates on stated assumptions; transaction fees are levers to be replaced with real schedules from production partners. Full derivations, per-sector growth tables, and sources are in the companion working notes *Rent and Shard Splitting* and *Sector Economics and Go-to-Market Viability*. · **F1R3FLY Industries**